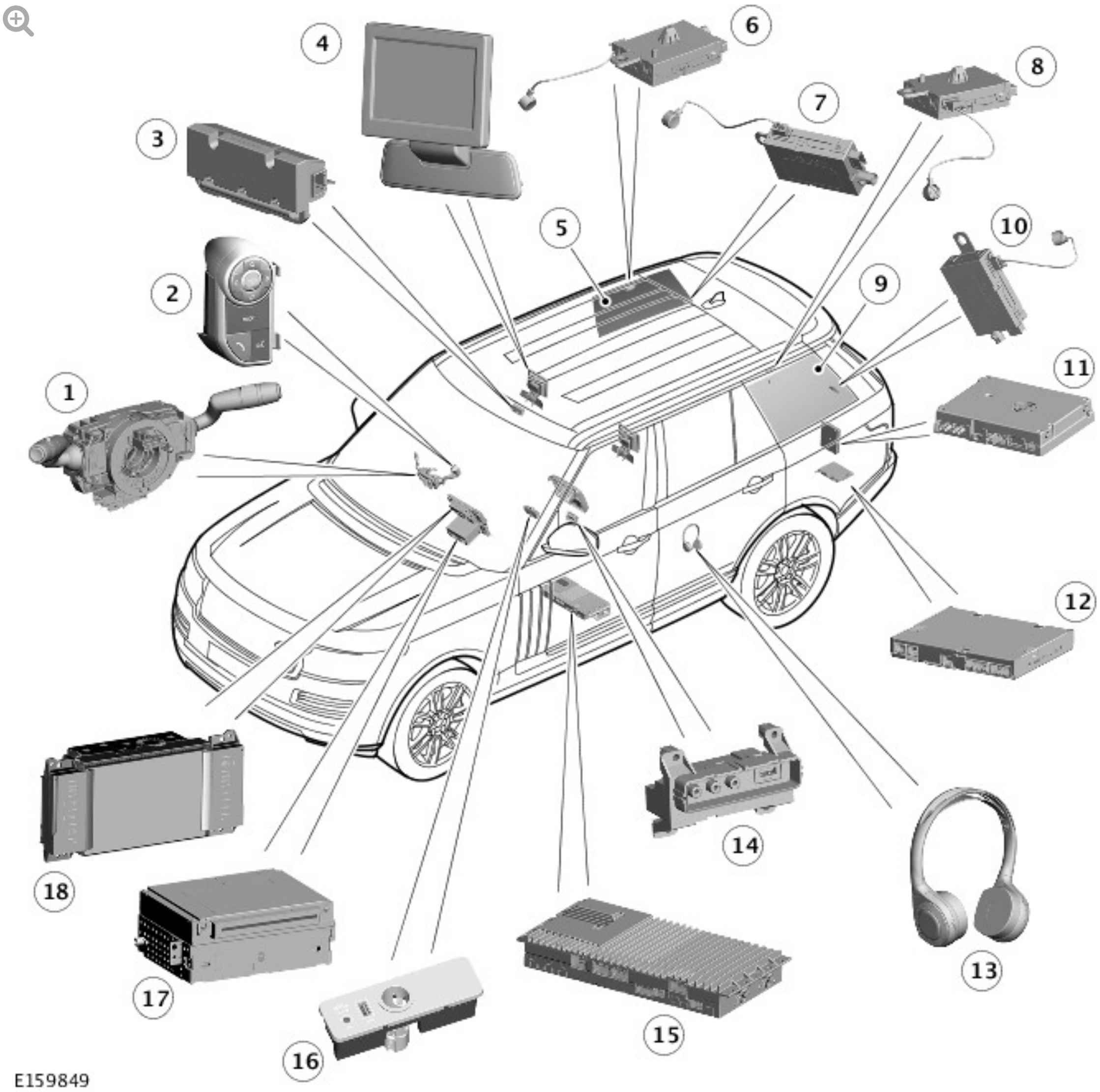
INFORMATION AND ENTERTAINMENT SYSTEM

DESCRIPTION AND OPERATION

COMPONENT LOCATION - VIDEO SYSTEM



ITEM	DESCRIPTION
1	Clockspring
2	Left steering wheel switchpack
3	Headphones transmitter
4	Rear Seat Entertainment (RSE) screen
5	Right rear quarter window TV antenna
6	Right upper Television (TV) antenna amplifier
7	Right lower TV antenna / Fuel Fired Booster Heater (FFBH) remote control antenna amplifier (when fitted)
8	Left upper TV antenna amplifier
9	Left rear quarter window TV antenna
10	Right lower TV antenna amplifier
11	TV Control Module (TVCM)
12	RSE control module
13	Wireless headphones (up to 3 off)
14	Audio Visual Input/Output (AVIO) panel (standard class location shown)
15	Audio Amplifier Module (AAM)
16	Portable audio interface panel
17	Integrated Audio Module (IAM)
18	Touch Screen (TS)

OVERVIEW

The fiber optic, Media Orientated System Transport (MOST) based system provides video and audio entertainment for the front and rear seat occupants. The system allows rear seat occupants to view DVD (digital versatile disc) video and TV on two RSE screens, listen to audio output via the vehicle speakers or wireless headphones or display video images on the RSE screens from an external source, such as a video player or games console.

TV and DVD video images can also be displayed on the Touch Screen (TS) if the vehicle is below a predetermined speed threshold or has dual-view TS fitted (where market regulations allow).

The system comprises the following components:

- RSE control module
- TV control module
- Four TV Antennas
- Four TV Antenna amplifiers
- Two RSE screens
- RSE remote control
- Wireless headphones
- AVIO panel.

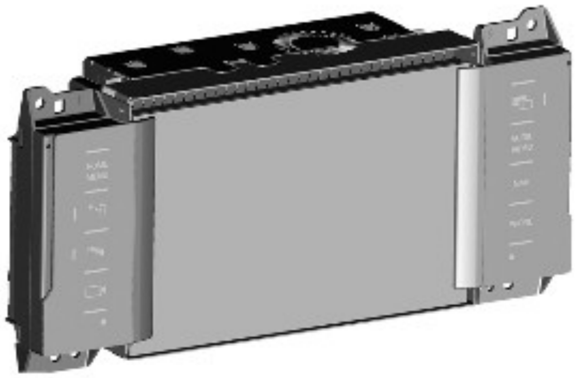
The RSE system also uses other components which form part of the audio system as follows:

- Touch Screen (TS)
  For additional information, refer to: Audio System (415-01 Information and Entertainment System, Description and Operation).
- Integrated Audio Module (IAM)
  For additional information, refer to: Audio System (415-01 Information and Entertainment System, Description and Operation).
- Audio amplifier module
  For additional information, refer to: Audio System (415-01 Information and Entertainment System, Description and Operation).
- Headphones transmitter
- Vehicle speakers.
  For additional information, refer to: Speakers (415-01 Information and

Entertainment System, Description and Operation).

DESCRIPTION

TOUCH SCREEN (TS)



E141842

The Touch Screen (TS) is located in the center of the instrument panel and is the driver control interface for the infotainment system. The TS comprises an 8 inch color, touch sensitive display, with two separate touch sensitive TS switchpacks located on either side. The TS is connected to the MOST ring and communicates with the other components in the audio/infotainment system.

The TS communicates with the RSE control module and TV control module via MOST ring. Composite Video Baseband Signal (CVBS) is used to transmit video images from/to the RSE control module and TV control module.

The TS also provides driver display and control of the audio system, telephone, the rear view camera, proximity cameras, and the navigation system.

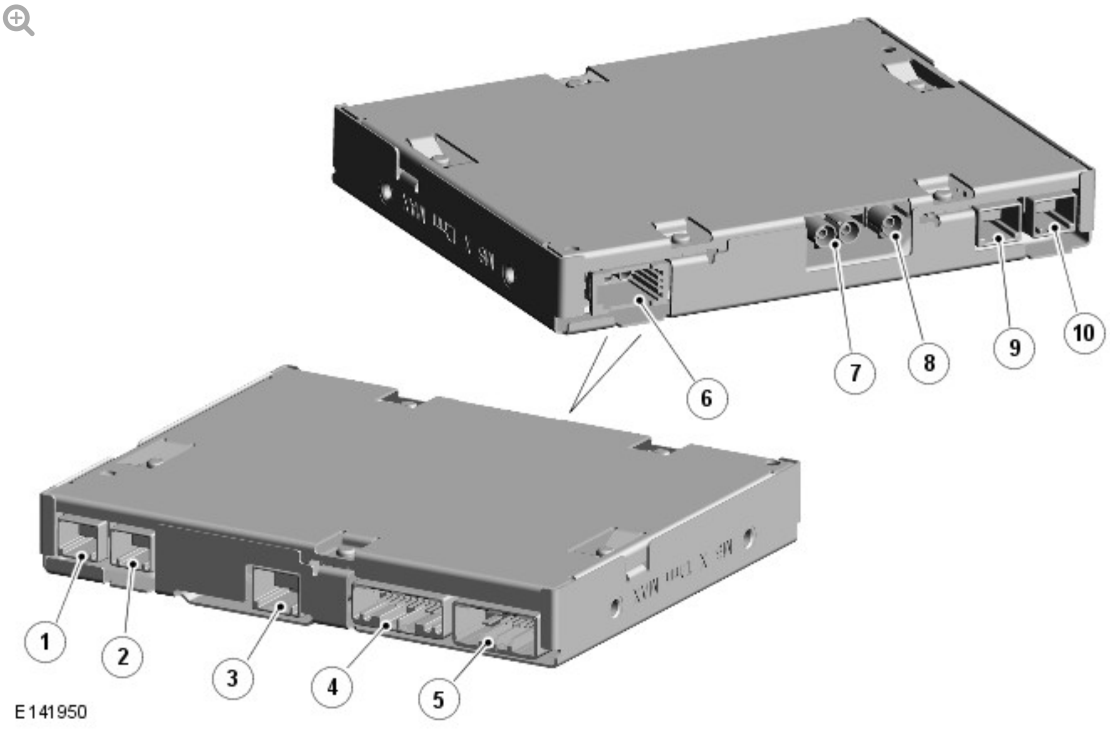
Two versions of the TS are available; single view and dual view. Additionally, TS switchpacks differ depending on system specification; for example, if Park Assist, navigation, dual view or rear view camera is specified on the vehicle, these soft keys will replace Setup, Audio Settings, Mode and display on/off respectively. The functions can still be accessed via the home menu on the TS.

NOTE:

Due to legislation, certain markets do not receive this dual-view option. A single view display is available in these markets.

For additional information, refer to: Audio System (415-01 Information and Entertainment System, Description and Operation).

REAR SEAT ENTERTAINMENT (RSE) CONTROL MODULE



ITEM	DESCRIPTION
1	LVDS connector - video output to left RSE screen
2	LVDS connector - video output to right RSE screen
3	AVIO panel connector - USB input
4	Connector - AVIO panel audio/video input, private medium speed CAN to RSE screens, RSE remote control data
5	Connector - 12V power supply to RSE screens, 12V power supply to RSE remote

	control docking station, power supply from RJB and ground
6	Connector - MOST
7	CVBS connector - TV video output to Touch Screen (TS)/Video input from TS
8	Not used
9	Not used
10	LVDS connector - video input from TV control module

The RSE control module is located in the rear left side, on the floor of the luggage compartment, behind the trim panel. The control module is attached to 2 brackets which are secured to the floor pan with 2 bolts and the rear side quarter panel with a nut on a captive stud.

The RSE control module manages the request signals from the RSE remote control, receives video and audio inputs from the IAM, TV control module and AVIO panel and outputs TV video to the TS and RSE screens and audio to the audio amplifier module.

The RSE control module is connected directly to the following modules for the purpose of processing audio, video, input and output signals:

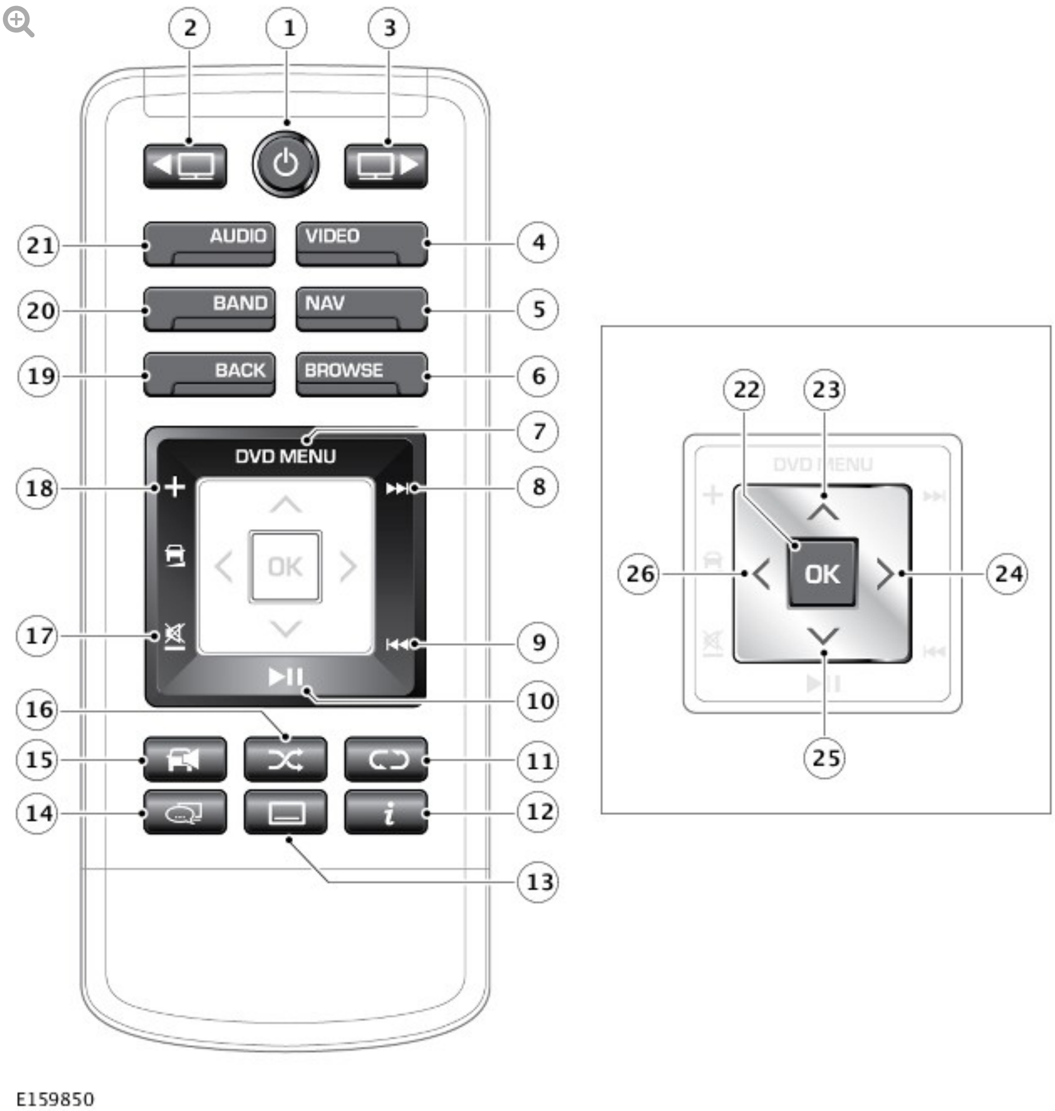
- TS - Connected via CVBS video - Processes DVD video from IAM to RSE screens
- Audio amplifier module - Connected via MOST - Processes audio signals for output on vehicle speaker system or headphones
- TV control module - Connected via Low Voltage Differential Signaling (LVDS) - TV signals are processed and passed to TS and RSE screens
- Rear screens/RSE remote control - Connected to RSE screens via LVDS and medium speed comfort CAN (controller area network) - Processes wireless infra-red signals
- AVIO panel - Connected via audio/video/USB (universal serial bus) - Processes audio and video signals from remote source.

The RSE control module receives a permanent power supply via a fused connected from the BJB (battery junction box) and the RJB (rear junction box).

REAR SEAT ENTERTAINMENT (RSE) REMOTE CONTROL

NOTE:

To prevent accidental damage, always store the remote control in the stowage area provided when not in use.



E159850

ITEM	DESCRIPTION
1	Press and release to power on/off screen
2	Left screen select

3	Right screen select
4	Video source select. Navigation summary. Japan and Brazil only: 1/12 screen segment select
5	Navigation summary. Japan and Brazil only: 1/12 screen segment select
6	Browse/TV channel list
7	DVD menu
8	Next track/file/channel/station
9	Previous track/file/channel/station
10	Play/Pause for all non-live media (USB/iPod/CD/DVD/Data Disc)
11	Repeat. Japan and Brazil only: Repeat and 'Red' selections
12	Information. Japan and Brazil only: Information and 'Yellow' selections
13	Subtitles on/off. Japan and Brazil only: Subtitles on/off and 'Green' selections
14	Audio/Video stream selection
15	Cabin audio volume on/off
16	Shuffle. Japan and Brazil only: Shuffle and 'Blue' selections
17	Cabin audio volume decrease/mute
18	Cabin audio volume increase
19	Up 1 level
20	Radio band select. Japan and Brazil only: Radio band select and interactive TV select
21	Audio source select Left screen select.
22	Select.
23	Cursor control: Up.
24	Cursor control: Right.
25	Cursor control: Down.
26	Cursor control: Left.

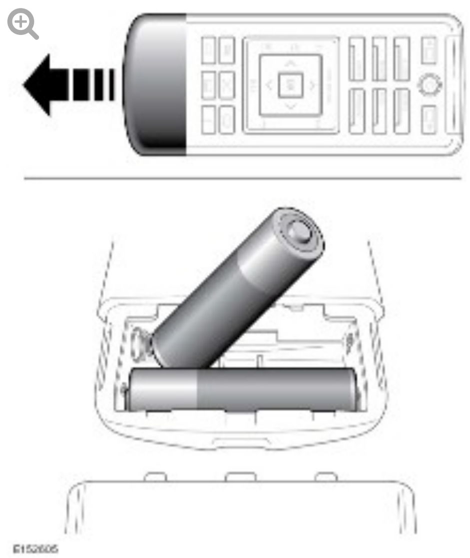
The RSE remote control has two locations dependant on vehicle specification; on standard vehicles it is located in the rear seat center armrest, on business class vehicles the it is located in the rear floor console.

The RSE remote control allows independent multimedia control for left and right rear seat passengers.

The RSE remote control is a has a number of switches to control the audio/video functions.

The RSE remote control allows selection of various entertainment system functions. When selected, activate menus displayed in the RSE screens mounted in the head restraints. The menus can be navigated using a five-way switch on the RSE remote control. For example, the user can select and press a soft key on the touch screen to activate a list of available radio stations in the RSE touch screen and then use the five-way switch to browse the list and select a radio station.

REAR MEDIA REMOTE CONTROL BATTERY



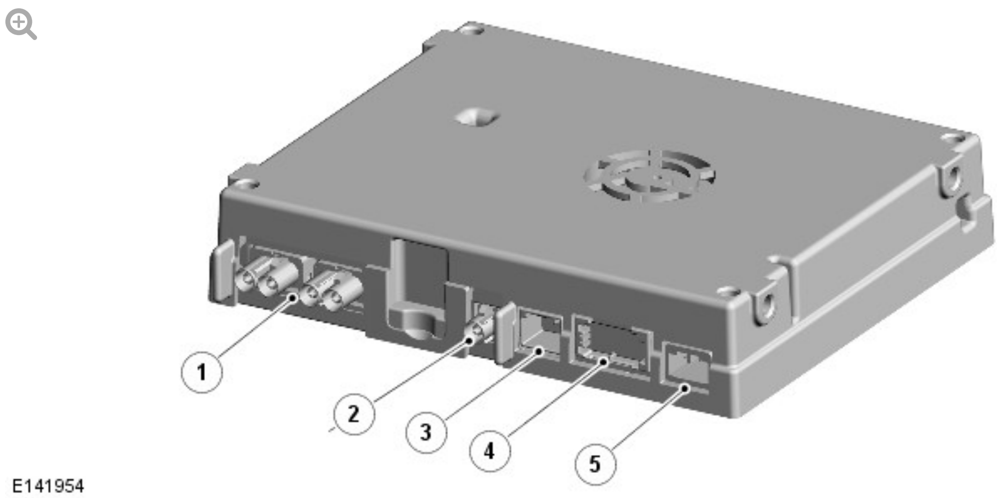
The RSE remote control is powered by 2 AAA batteries located inside the chrome cover. Low battery power is indicated by the remote control button flashing 3 times when pressed.

The RSE remote control transmits an infra-red digital signal in response to operation of a switch or soft key. The infra-red signal is received by a receiver sensor located on each RSE screen. The RSE remote control also allows selection of an auxiliary input from the AVIO panel (video or games console) or selection of audio (radio or CD (compact disc)).

The RSE remote control can be used to control functions including radio, CD/ DVD, plug-in audio devices, and TV (television), by displaying options in the touch screen, which activate menus displayed on the RSE screens in the head restraint, which are navigated using the five-way switch on the RSE remote control. For example, the user can press a soft key on the RSE remote control touch screen to activate the list of available radio stations in the RSE screen display. The user can then operate the five way switch to browse the list and select from it.

When the RSE remote control is used out of the docking station, the infra-red transmission takes place between the RSE remote control and the RSE screens. User requests are passed onto the RSE control module to be processed via a private medium speed CAN link.

TELEVISION (TV) CONTROL MODULE



ITEM	DESCRIPTION
1	Connectors - Antennas
2	CVBS connector - to Touch Screen (TS)
3	LVDS connector - to RSE control module
4	Connector - MOST

5	Connector - power from Rear Junction Box (RJB) and ground

The TV control module is at the rear of the luggage compartment, in the spare wheel well. The TV control module is connected to the infotainment system using one harness connector, one LVDS connector (for RSE vehicles), one single FACRA connector for CVBS connection to the TS (non RSE vehicles), one MOST connector and two dual FACRA connectors for the antenna connections.

On vehicles with RSE, the video output is passed from the TV control module on LVDS to the RSE control module and from the RSE control module on CVBS to the TS. On a vehicle with RSE fitted the CVBS output from the TV module is not used.

On vehicles without RSE, the video output is passed from the TV control module on CVBS to the TS. LVDS output is not used.

The front seat occupants can view the TV transmissions on the TS if the vehicle is stationary (market dependent). If the vehicle is fitted with a dual view TS then the front passenger can view the TV transmissions even when the vehicle is moving, but the driver will only be able to watch TV transmissions (or video) when stationary. This can vary depending on local market laws. When the RSE system is fitted, the TV control module allows the rear seat occupants to view television transmissions on the RSE screens.

Due to analogue transmissions Worldwide being switched off, Land Rover are introducing more TV control modules to support the numerous digital TV standards that are being implemented. To ensure that analogue and digital transmissions are received in all markets, three types of TV control module are available. The type of TV control module fitted is dependant on which digital standard is supported in different markets (if at all).

- Digital Hybrid TV control module - this TV control module supports digital DVB-T standard (MPEG2 only) and analogue TV transmissions. This TV control module is only fitted to markets that still have significant analogue broadcasts and minor digital coverage or to vehicles in markets that still have significant analogue broadcasts and where Land Rover do not currently have a TV control module that supports the digital TV standard in that market. This TV control module requires the use of a link lead that combines the power/ground and CVBS output into one connector that mates with the connectors on this type of module.

- DVB-T TV control module - this TV control module is digital only (no analogue tuner) and supports both MPEG2 and MPEG4 digital compression standards. This TV control module is fitted in any market that has significant DVB-T coverage (most of Europe and some of the rest of the World).
- ISDB-T TV control module - this TV control module is for the Japanese market only and receives ISDB-T digital broadcasts only (no analogue). This TV control module requires the use of a link lead that combines the power/ground and CVBS output into one connector that mates with the connectors on this type of module.

The TV control module is connected on the MOST ring which it uses to output its audio signals to the audio amplifier module. Video output from the TV control module is on an LVDS cable to the RSE control module or a CVBS screened co-axial cable to the TS in vehicles without the RSE system. Four further connections provide for the signal input from four TV antenna amplifiers which are in turn attached to four antennas in the rear quarter windows.
For additional information, refer to: Antenna (415-01 Information and Entertainment System, Description and Operation).

The TV control module contains three or four internal tuners (depending on the TV control module fitted); all of the tuners are connected to the antennas. The tuners receive the audio and video signals. The tuner, or combination of tuners with the strongest signal, are automatically used to display the required TV channel.

One of the internal tuners is always used to scan the locality for receivable channels (background scanning). The tuner can detect different frequencies transmitting the same channel and can select the strongest signal for use.

In certain areas signal strengths will vary. When in an area of weak reception, a break-up in picture and sound quality, or a blank screen or frozen picture and audio muting may occur. Dependant on the type of TV control module fitted, it may be of benefit to switch between analogue and digital TV stations.

The main advantage of digital TV reception is the improved picture quality. Four TV antennas are located in the rear quarter windows. In the event of a loss of digital reception, a 'loss of reception' message is displayed. The TV system also has a programme diversity function as well as antenna diversity that allows the current tuned programme to be maintained as the vehicle passes through regions with different transmitters broadcasting on different frequencies. This function is only occurs where the reception data for the presently tuned channel matches the data of the new signal.

The system offers a choice of aspect ratios, between 4:3, 16:9 and zoom. The zoom ratio will always try to fill the screen and avoid the black bars/squashed image. For digital signals the broadcaster will usually send the correct format for the transmitted programme and this will remove one of the incorrect formats from the screen (4:3 or 16:9). The "Zoom" option is always available and will always be selected as the default format. If the format is unknown (typical for analogue) then all three format options will be displayed. The TS offers picture quality using a resolution of 800x480 pixels for single view and 400x480 pixels for dual view.

TV CONTROL MODULE - JAPAN

Japanese market vehicles are fitted with a TV control module unique to that market. The control module has a slot to allow a B-CAS (BS Conditional Access Systems Co., Ltd.) card to be inserted. All digital TV's in Japan (home systems included) require a conditional B-CAS access card. This card decrypts the TV broadcast signal to allow it to be displayed as all broadcasts in Japan are encrypted. Without this card there will be no picture or audio.

B-CAS (BS Conditional Access Systems Co., Ltd.) is a vendor and operator of the Integrated Services Digital Broadcasting (ISDB) CAS system in Japan.

All ISDB receiving apparatus requires a B-CAS card under regulation, the B-CAS card is supplied as the standard accessory in Japan.

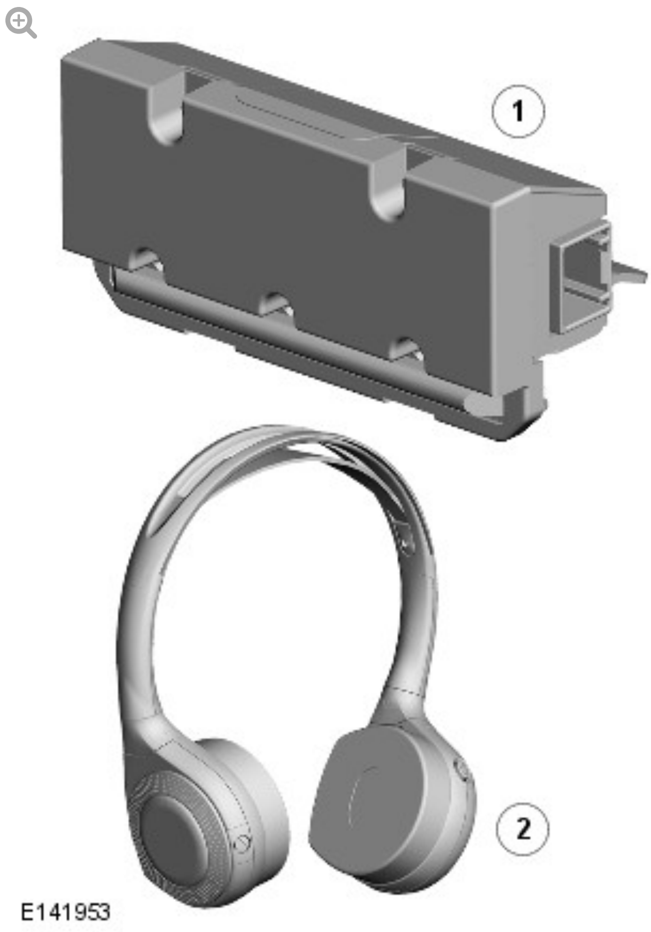
Integrated Services Digital Broadcasting (ISDB) is a Japanese standard for digital television and digital radio used by the country's radio and television stations.

TELEVISION (TV) ANTENNAS

TV antennas are located in the rear side windows. Refer to the Antennas section.

For additional information, refer to: Antenna (415-01 Information and Entertainment System, Description and Operation).

WIRELESS HEADPHONES



ITEM	DESCRIPTION
1	Headphones transmitter

The headphones transmitter is located in the roof headliner near the front overhead console assembly.

The headphones transmitter transmits audio output for reception by the wifeless headphones for the front and rear seat passengers. The headphone transmitter is connected to the audio amplifier module via a Low Voltage Differential Signal (LVDS) plug connector. Audio data is relayed to the headphone transmitter from the audio amplifier module via the LVDS line. The audio amplifier module also supplies the power and ground for the headphones transmitter. The audio signal, is then passed on to the wireless headphones via a Whitefire® digital infra-red signal to any seating position within the interior of the vehicle. The digital signal means that each headphone user is able to listen to a different audio source.

NOTE:

> Providing an audio source is directed to the headphones transmitter, faint red lights can be seen behind the smoked glass cover of the transmitter. This can be useful for diagnosis to prove audio infra-red transmission is active.

The headphone system includes Dolby® headphone surround sound when listening to the DVD source.

NOTE:

> There is no docking station to store the wireless headphones, therefore charging of the batteries is not supported via the vehicle electrical system.

The system can support up to three pairs of wireless headphones. When a dual-view screen is fitted, the front passenger can listen to TV/DVD (digital versatile disc) and when Rear Seat Entertainment (RSE) screens are fitted, rear passengers can listen to selected source on RSE screen. The headphones have an adjustable headband which operates on a ratchet mechanism.

The left side and the right side of the wireless headphone houses the infra-red receiver sensors which receive the transmitted signals from the headphones transmitter. Two AAA batteries are located below a sliding cover. When inserting the batteries it is important that the battery polarity is observed as marked in the battery compartment.

The right side of the wireless headphone houses the volume control, a channel switch and a power 'ON' LED (light emitting diode). The channel switch allows the user to select alternative audio channels (rear left/right, dual-view audio source) when active. The power 'ON' LED is illuminated when the on/off switch on the right side of the wireless headphone is operated. The LED will remain on and the wireless headphones powered until the switch is operated for a second time. If the wireless headphones have not received an infra-red signal from the headphones transmitter for several minutes, they will automatically switch off to prevent battery drain.

REAR SEAT ENTERTAINMENT (RSE) SCREENS

E157958

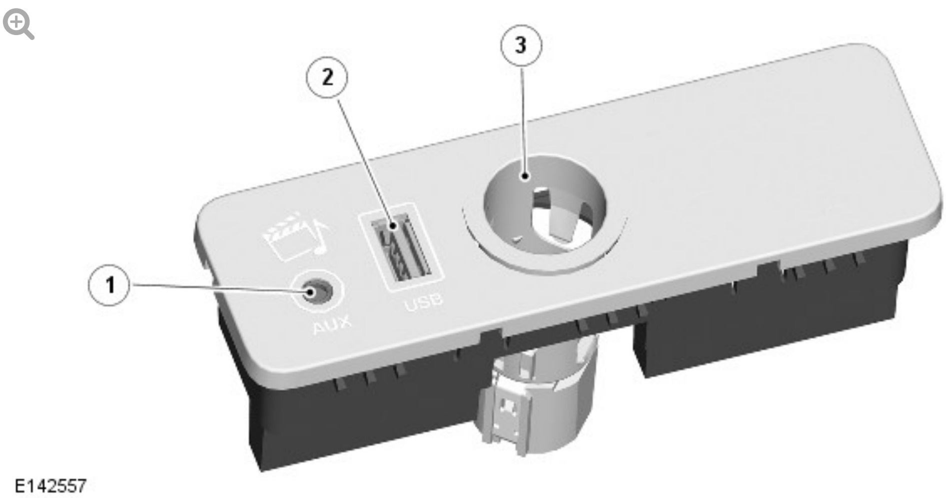
The RSE screens are located in the rear of the front seat head restraints. The screen is secured in the head restraint with one screw and 2 metal clips which are covered by a removable surround. The screen is an 8 inch, auto dimming (selectable via RSE remote control settings), 800X480 resolution Thin Film Transistor (TFT) monitor.

An infrared receiver sensor is located centrally in the bottom portion of the screen surround. The sensor receives infrared transmissions from the RSE remote control and passes them to the RSE control module on a private CAN bus system which is only connected between the RSE control module and the RSE screens. The RSE control module can then transmit any relevant messages onto the MOST ring. All screen settings can be changed using the RSE remote control.

The screen should be cleaned with a lightly, water moistened cloth. Do not use chemical agents or domestic products to clean the screen or any part of the surround.

Each RSE screen is connected to the RSE control module using a 12 pin harness connector and 2 pin LVDS video connector.

PORTABLE AUDIO INTERFACE PANEL



ITEM	DESCRIPTION
1	3.5mm jack plug - MP3 connection
2	Universal Serial Bus (USB) connector
3	12V accessory socket

The portable audio interface panel is located in the floor console below the stowage box lid and is fitted to all audio systems. The interface is a media hub between a portable input device and the IAM. The interface contains two USB ports, an iPod® port and a 3.5mm jack plug connector.

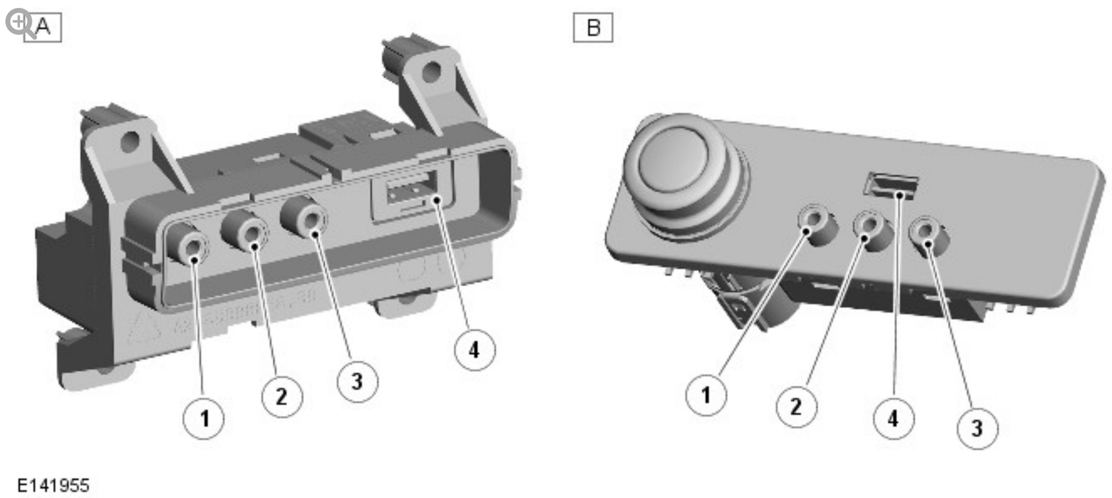
The portable audio interface panel is connected to the vehicle with two harness connectors. The harness connector from the USB ports has gold plated pins.

Devices that can be connected to the portable audio interface panel include:

- USB mass storage devices (for example a memory stick). Devices must use FAT or FAT32 file format.
- iPod® (iPod Classic, iTouch, iPhone and iPod Nano are supported - full functionality for older devices cannot be guaranteed). iPod Shuffle functionality cannot be guaranteed.
- Auxiliary device (personal audio, MP3 players).
- Devices with Bluetooth® connectivity. Devices must support A2DP and AVRCP Bluetooth® protocols).

For additional information, refer to: Audio System (415-01 Information and Entertainment System, Description and Operation).

AUDIO VISUAL INPUT/OUTPUT (AVIO) PANEL



ITEM	DESCRIPTION
A	Standard AVIO panel - located in rear of floor console
B	Business class AVIO panel - located in rear seat console
1	Yellow socket - Composite video input
2	White socket - Left audio input
3	Red socket - Right audio input
4	USB

The AVIO panel is located in the rear of the floor console below the rear Integrated Control Panel (RICP) on standard vehicles and in the rear seat console, below the armrest on business class vehicles.

The panel provides for the connection of auxiliary audio and video inputs from an external source, such as a games console, via three AV sockets on the panel.

The sockets are connected to the RSE control module and allow the auxiliary input video to be played on the RSE screens and the audio to be played on the vehicle speakers or on the wireless headphones. The auxiliary input video cannot be displayed on the TS.

The single USB plug allows for the attachment of an auxiliary audio input, such as iPod®/iPhone®, an MP3 player or a USB memory stick. This plug is connected directly to the RSE control module and allows audio to be selectable on the RSE screens and played on the wireless headphones or through the vehicle speakers. It will also allow standard definition Div X video files to be played from a USB memory stick.

The AVIO panel is connected to the RSE control module using a single 20 pin harness connector.

OPERATION

TELEVISION

The television system has various levels of user control through the Touch Screen (TS), the Integrated Control Panel (ICP) and the left steering wheel switchpack. The system includes 6 analogue and 12 digital channel pre-sets. As with the audio system, the user can tune up or down through the channels and store a channel by a long press of the selected preset button. The system offers a choice of screen aspect ratios.

For the digital bands, the options available on the TS are the correct aspect ratio for the broadcast and "zoom" which will fit the picture to the screen (no black bands). For analogue TV all aspect ratio options will be available including 16:9, 4:3 and zoom. Two antennas are used because the system is dynamic and each one is connected to an individual tuner. The TV control module evaluates which antenna has the strongest signal and will use a combination of antennas to generate the TV image and sound. At any point in time the TV control module is using either one or two of the antennas and it's tuners to scan the TV frequencies and generate an up to date channel list.

The television system is primarily controlled from the TS and the ICP which are located in the center of the instrument panel and also the left steering wheel switchpack. Control signals from the TS are sent on the Media Oriented System Transport (MOST) ring to the TV control module. The TV control module uses a dedicated Composite Video Baseband Signal (CVBS) bus to transmit video signals to the TS. Where RSE is fitted, the TV can also be controlled using the RSE remote control and the video output will be delivered to the RSE control module via an Low Voltage Differential Signalling (LVDS) video link. The TV video feed is then forwarded on to the TS from the RSE control module via a CVBS output.

Control signals from the ICP and the left steering wheel switchpack are relayed on the medium speed comfort CAN bus to the TS. The TS relays the control signals to the TV control module on the MOST ring. The TS is the bus master for the MOST ring and also hosts a gateway function between the medium speed comfort CAN bus and the MOST ring.

The TV control module audio output signals are sent on the MOST ring to the audio amplifier module for speaker output.

REAR SEAT ENTERTAINMENT (RSE)

The RSE control module is connected directly to both rear RSE screens via a medium speed comfort CAN link and a Low-Voltage Differential Signaling (LVDS) connection. Infra-red signals from the RSE remote control are received by the RSE screens and user requests are communicated via a private medium speed CAN link to the RSE control module. Video signals are communicated via the LVDS to one or both screens as requested.

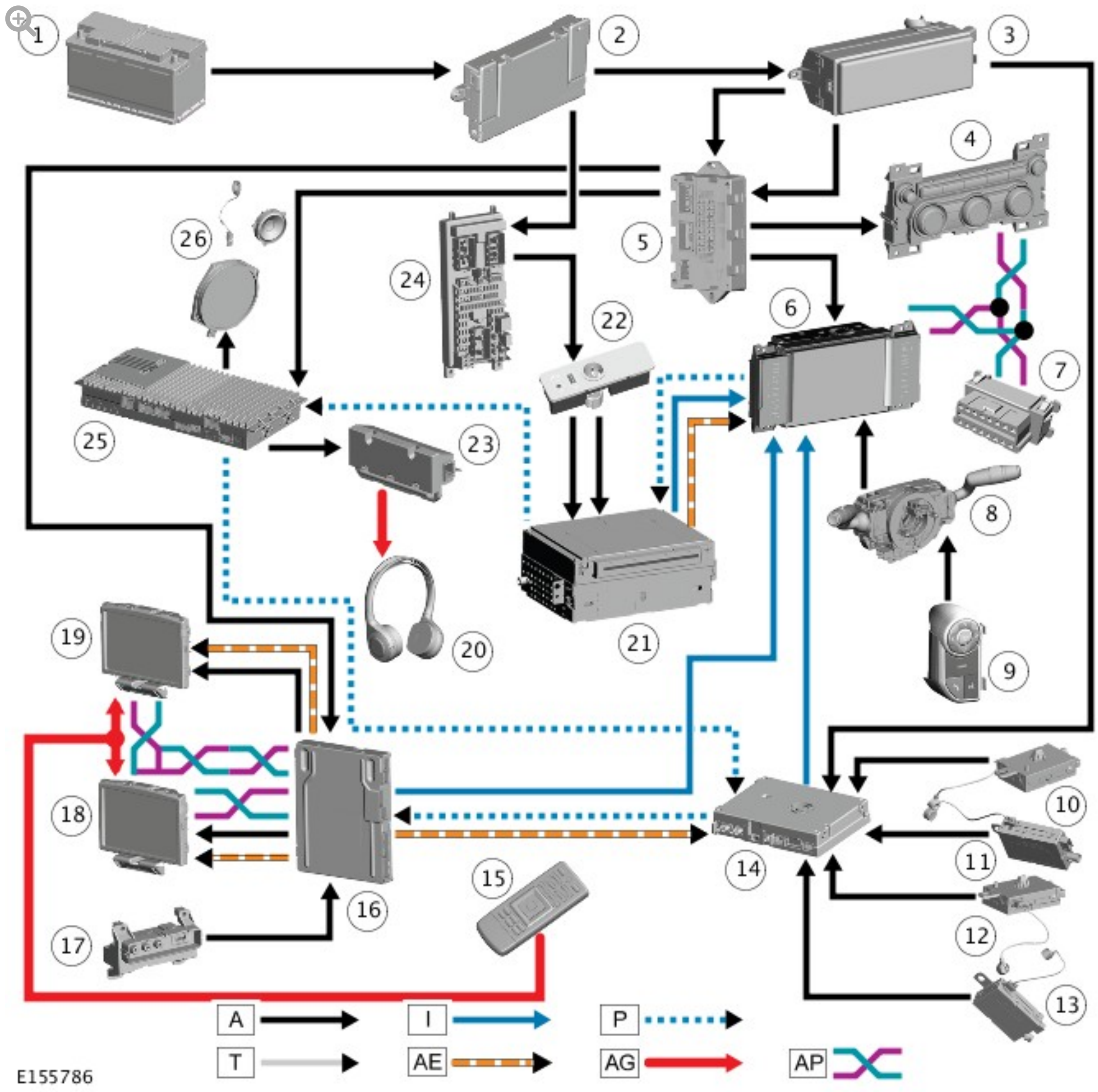
The RSE control module communicates with the audio system via the MOST connection. Audio input from the Audio Video Input/Output (AVIO) panel is processed by the RSE control module and passed on the MOST ring to the audio amplifier module to allow audio output to be played on the vehicle speakers or on the wireless headphones.

Video input from the TV control module and the AVIO panel is also processed by the RSE control module and passed to the two RSE screens. The TV Video is also passed to the TS on separate video connection. The RSE control module also controls the power supplies to the RSE screens and relays the infra-red RSE remote control signals received by the RSE screen infra-red sensors. The infra-red signals are passed from the RSE screens to the RSE control module on a local CAN bus system. The navigation system route can also be displayed on the RSE screens
For additional information, refer to: Navigation System (415-01 Information and Entertainment System, Description and Operation).

.

A time limit operation is active when the ignition is off and the vehicle is unlocked (Power mode 4) and the system is manually switched on using the TS. The system will operate for a maximum of ten minutes. The battery voltage is continually monitored by the IAM. If the IAM detects that the battery voltage has fallen to a predetermined level, the IAM will shut the infotainment system down to prevent further battery drain. Once the system has shut down due to low battery voltage, it can only be restarted when the engine is running and the battery voltage has risen above the threshold level for more than one minute.

CONTROL DIAGRAM



A = HARDWIRED; F = RF (RADIO FREQUENCY) TRANSMISSION; I = CVBS (COMPOSITE VIDEO BASEBAND SIGNAL); AP = MS (MEDIUM SPEED) CAN COMFORT BUS; P = MOST (MEDIA ORIENTED SYSTEM TRANSFER); T = COAXIAL; AE = LVDS (LOW-VOLTAGE DIFFERENTIAL SIGNALLING); AG = INFRARED

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB)

3	Rear Junction Box (RJB)
4	Integrated Control Panel (ICP)
5	Quiescent Current Control Module (QCCM)
6	Touch Screen (TS)
7	Diagnostic socket
8	Clockspring
9	Left steering wheel switchpack
10	Right upper TV antenna amplifier
11	Right lower TV antenna amplifier/Fuel Fired Booster Heater (FFBH) remote control antenna amplifier (when fitted)
12	Left upper TV antenna amplifier
13	Left lower TV antenna amplifier
14	TV Control Module (TVCM)
15	RSE remote control
16	RSE module
17	AVIO panel
18	Right RSE screen
19	Left RSE screen
20	Wireless headphone (3 off)
21	Integrated Audio Module (IAM)
22	Portable audio interface panel
23	Headphones transmitter
24	Central Junction Box (CJB)
25	Audio Amplifier Module (AAM)
26	Vehicle speakers